

OLUBUNMI OLARINDE

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with a strong foundation in statistical modeling, NLP, computer vision, and cloud ML systems. Combines hands-on modeling and deployment experience (PyTorch, scikit-learn, AWS SageMaker, Hugging Face) with technical program management at Microsoft. Passionate about building human-centered, responsible-AI systems that are technically rigorous, fair, and grounded in the experiences of the people they serve.

TECHNICAL SKILLS

Languages: Python, R, SQL, HTML, CSS

ML / AI: GenAI/LLM pipelines, RAG, FAISS, NLP, computer vision, deep learning, supervised/unsupervised learning; PyTorch, scikit-learn, TensorFlow/Keras, Hugging Face Transformers, Sentence Transformers, Gensim, TF-IDF, TextBlob, llama.cpp

Data & Analytics: EDA, statistical modeling, experimentation/evaluation, feature engineering, A/B testing, data cleaning, data integration, business insights

Visualization & BI: Power BI, Tableau, Plotly, Matplotlib, Seaborn; reporting and insight storytelling for stakeholder decisions

Cloud / MLOps: AWS (S3, SageMaker, Lambda, Step Functions, EC2, IAM, Model Monitor), Azure Logic Apps/Azure OpenAI exposure, Azure (fundamentals), Hugging Face Hub/Spaces, Streamlit, FAISS, Docker (basic), Git/GitHub

PM / Product / Collaboration: Requirements scoping, validation project plans, AI scoring rubrics, go/no-go checklists, stakeholder communication, status reporting, cross-functional coordination, Scrum, ADO, technical documentation; Microsoft 365, Copilot, Teams, SharePoint, Excel, Power BI, Jupyter

EXPERIENCE

Technical Program Manager Fellow - Microsoft

May 2026 - Present

- Led a production AI-assisted email-response workflow that processed 1,000+ messages in 10 weeks and reduced manual review volume by nearly 90% through automated triage.
- Maintained 99%+ critical-failure-free operation across 1,000+ production runs by implementing monitoring, automated recovery, incident runbooks, and a 30+ case regression suite.
- Achieved 90%+ first-pass approval for AI-generated drafts by grounding responses in approved knowledge sources and surfacing source-coverage signals for reviewers.
- Eliminated shared credentials and duplicate customer replies by designing a secure human-in-the-loop approval workflow using managed identity and asynchronous messaging.
- Prepared an AI-assisted review workflow for handoff by delivering integration plans, design documentation, a four-dimension scoring rubric, a seven-case test matrix, and go/no-go criteria.

Faculty Teaching Assistant - (AI Ethics) - Eastern University

Nov 2025 - Present

- Scaled AI Ethics capstone support across 3-4 rolling 7-week cohorts of 200-250 graduate students by serving as assessment lead for ~100-125 students per cohort within a 7-20 hr/week workload.
- Maintained a 24-48 hr student-support SLA while assessing 300+ AI Ethics artifacts per cycle (15-20 min videos, scripts, decks, ≥10 cited sources) and applying full accessibility accommodations.
- Sustained A/B median pass rates across cohorts by converting responsible-AI expectations into structured rubric feedback, one-on-one revision guidance, and clear documentation standards.
- Improved practitioner readiness on 3 core responsible-AI concepts (fairness, transparency, responsible data use) by translating complex technical and ethical material into concrete capstone guidance for graduate-level students.

Founder & Lead ML Engineer - Smart Sickle Cell Diary — Independent Research Jun 2025 - Nov 2025

- Why: Sickle Cell Disease affects ~7.74M people globally (Lancet GBD 2021) and ~100K in the US (CDC), disproportionately African, Caribbean, and other marginalized communities; daily symptom logging tools remain clinical and burdensome.
- v1 (modeling): Built a Streamlit + Hugging Face Spaces logger with a predictive backbone over a 3,000-log, 25-persona synthetic dataset (Azure Health Bot); engineered TF-IDF + TextBlob sentiment features; benchmarked 7 regressors and 5 emotion classifiers. Feature-based upsampling on binned age + sentiment cut Pain MAE 2.70 → 1.29 (R^2 -0.29 → 0.53) and lifted Emotion Accuracy 34.7% → 82.9% (macro-F1 0.34 → 0.82).

- Responsible-AI pivot to v2: Re-scoped the system after recognizing that predicting symptoms for patients risked minimizing lived experience; pivoted to amplify patient voices via a modular FAISS-backed RAG architecture (streamlit_app.py, rag_integration.py, trigger module) over a curated SCD corpus, surfacing supportive Insight Cards with citations, transparent user controls, and CSV export.
- Validation: Ran a 14-day mini-pilot with real testers on de-identified logs; improved onboarding via how-to video and in-app tips. Code: huggingface.co/spaces/olubunmiamoke/scd_mini_pilot
- Next: Expand RAG corpus, participatory co-design with patient communities, multilingual coverage, clinician dashboard, IRB-track validation.

SELECTED PROJECTS

Sentiment Analysis of Stock-Related News (Python, Sentence Transformers, Gensim, llama.cpp, Hugging Face Hub)

- Selected the strongest NLP approach for NASDAQ-equity news by benchmarking 6 model variants (Word2Vec, GloVe, Sentence Transformer × base + tuned), with Sentence Transformer baseline performing best (val Acc 0.62 / F1 0.50).
- Delivered weekly decision-ready equity summaries by building a quantized Llama pipeline via llama.cpp + Hugging Face Hub that emitted structured JSON of the top 3 positive and top 3 negative events per equity for downstream use.

Landmark Classification with CNNs (PyTorch, TorchVision, TorchScript, OpenCV, Pillow, pytest)

- Improved landmark classification accuracy by +34 pts by fine-tuning ResNet50 to 80% test accuracy versus a 46% scratch CNN baseline across 50 global landmarks and 1,250 test images.
- Operationalized a top-5 image-classification demo by designing a 4-conv-block architecture with batch norm + dropout, label-smoothed cross-entropy, Adam, RandomAffine/ColorJitter/HorizontalFlip on 224×224 inputs, pytest-validating the predictor, and deploying via TorchScript + ipywidgets.

Scones Unlimited - End-to-End ML on AWS (SageMaker, Lambda, Step Functions, S3, IAM, Boto3)

- Reached ~98% test-inference confidence by architecting a serverless AWS pipeline where Step Functions chained 3 Lambdas (S3 fetch + base64 serialization → SageMaker endpoint invoke → 0.93 confidence-threshold filter) behind a ml.m5.xlarge real-time endpoint.
- Created a reproducible 2-class deployment by training a bicycle-vs-motorcycle classifier on a CIFAR-100 subset with SageMaker's built-in image classifier and configuring IAM roles, S3 buckets, and endpoint scaling.
- Enabled 100% inference data capture for drift and endpoint-health monitoring by configuring SageMaker Model Monitor DataCaptureConfig to S3 JSONL and analyzing captures with jsonlines + Matplotlib.

EDUCATION

M.S., Data Science - Eastern University

Postgraduate Certificate, AI / Machine Learning - University of Texas at Austin

B.A., Music - Obafemi Awolowo University

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900); Azure AI Fundamentals (AI-901) - 2026
- Google Data Analytics Professional Certificate (Coursera) - 2026
- Udacity Nanodegree: AI Programming with Python / ML Fundamentals - 2024
- AWS & Udacity AI/ML Scholar - 2023-2024